GENETICS

Reading a Genetic Analysis Report

Genetics GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. What does the "c." mean at the start of a nucleotide change notation like c.1234C>T?
2. Name the three variant classification tiers covered in this lecture.
3. If a patient is heterozygous for one pathogenic gene variant, what does that mean for them and their family planning?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

This lecture is a walkthrough of how a clinical genetic test report is structured and read — not a disease lecture. The worked example (PKHD1, congenital hepatic fibrosis) is a template case used to illustrate the format.

1 Anatomy of a genetic test report

Section	What it contains
Patient info	Name, DOB, gender, sample ID, collection date, report date
Referring physician	Who ordered the test, institution, contact info
Test details	Test name, purpose, sample type (e.g. blood, saliva)
Methodology	DNA extraction method, sequencing method (NGS or Sanger), targeted genes, bioinformatics tools used
Results	Gene, nucleotide change, protein change, zygosity, variant classification, clinical significance
Clinical interpretation	What a positive, carrier, or negative result means in plain language
Limitations	What the test can't rule out
Recommendations	Referral, genetic counselling, further testing

2 Reading the nucleotide change notation

Worked example from the slides: **c.1234C>T**

c.	Describes the change at the **coding DNA sequence** level
1234	Position of the nucleotide in the gene
C	Original nucleotide at that position (Cytosine)
>	Indicates a substitution
T	New nucleotide replacing it (Thymine)

3 Variant classification

Three tiers were given in this lecture.

Classification	Meaning
Pathogenic	Confirmed to cause the disease; associated with increased risk
Likely pathogenic	Strong evidence it causes disease, but not yet enough data for a definitive call
VUS	**V**ariant of **U**ncertain **S**ignificance — insufficient evidence to classify as benign or pathogenic

4 Worked example — congenital hepatic fibrosis

Gene	**PKHD1**
Nucleotide change	c.1234C>T
Protein change	p.(Arg412Trp)
Zygosity	Homozygous
Classification	**Pathogenic**
Significance	Supports the clinical diagnosis; confirms a genetic basis for the patient's liver condition

5 Interpreting the three possible outcomes

Result	What it means
Positive	Mutation identified is associated with the disease — supports the clinical diagnosis and a genetic basis for the condition
Carrier	One pathogenic variant found (heterozygous) — doesn't cause disease in this person, but matters for family planning
Negative	No pathogenic variants found in the genes analyzed — reduces likelihood of the condition but **does not rule it out** (other causes still possible)

P Must-remember pearls

- **c.** = coding DNA position; the number = nucleotide position; letter>letter = the base substitution.
- Three classification tiers taught: **Pathogenic**, **Likely pathogenic**, **VUS**.
- Homozygous pathogenic variant = disease-causing in this patient; heterozygous pathogenic = carrier status.
- A negative result reduces likelihood of the tested condition but never fully rules it out.
- Every report ends in follow-up: referral to a specialist + genetic counselling + possible further testing.

Q Test yourself

1. A report lists a variant as "c.1234C>T." What does the "c." indicate?

A) A protein-level change
B) A coding DNA sequence-level change
C) A chromosomal location
D) Zygosity

2. A variant has strong evidence for pathogenicity, but not yet enough data for a definitive classification. How is it classified?

A) Pathogenic
B) Likely pathogenic
C) VUS
D) Benign

3. A variant’s clinical significance cannot be determined from current evidence. What is this called?

A) Pathogenic
B) Likely pathogenic
C) Variant of Uncertain Significance (VUS)
D) Polymorphism

4. A patient is found to carry one pathogenic PKHD1 variant (heterozygous). What is the most appropriate next recommendation?

A) Immediate liver transplant
B) Genetic counselling on carrier status and family planning
C) No further action needed
D) Repeat the same test in 6 months

5. A genetic panel returns negative for all analyzed genes in a patient with unexplained liver disease. What is the correct interpretation?

A) The patient definitely does not have a genetic liver disease
B) The likelihood of the tested condition is reduced, but other causes are not ruled out
C) The test was invalid
D) The patient is a confirmed carrier

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — B) Coding DNA sequence-level change — the "c." prefix in HGVS notation.

2 — B) Likely pathogenic — strong but not yet definitive evidence.

3 — C) VUS — insufficient evidence either way.

4 — B) Genetic counselling — carrier status matters for relatives and family planning, not for the patient's own disease management.

5 — B) Reduces likelihood but doesn't rule out other causes — a negative result is never fully exclusionary.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam